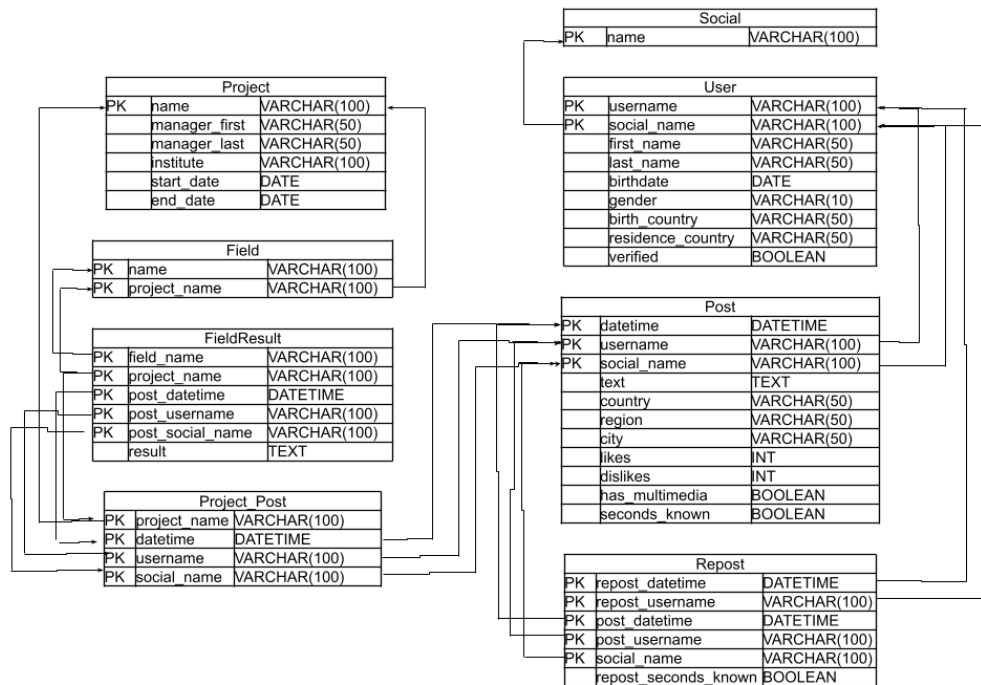


# Project Report: Social Media Analysis Database

## Database Schema



### Key to Foreign Keys

User: FOREIGN KEY (social\_name) REFERENCES social(name)  
Post: FOREIGN KEY (username, social\_name) REFERENCES user(username, social\_name) ON DELETE CASCADE  
Repost: FOREIGN KEY (repost\_username, social\_name) REFERENCES user(username, social\_name)  
FOREIGN KEY (post\_datetime, post\_username, social\_name) REFERENCES post(datetime, username, social\_name)  
Field: FOREIGN KEY (project\_name) REFERENCES project(name)  
FieldResult: FOREIGN KEY (field\_name, project\_name) REFERENCES field(name, project\_name)  
FOREIGN KEY (project\_name, post\_datetime, post\_username, post\_social\_name) REFERENCES project\_post(project\_name, datetime, username, social\_name)  
Project\_Post: FOREIGN KEY (project\_name) REFERENCES project(name)  
FOREIGN KEY (datetime, username, social\_name) REFERENCES post(datetime, username, social\_name)

### Table Usage:

**Social:** Stores names of social media platforms

**User:** Stores relevant user data; username and social\_name are required. Each user must be on a relevant social media platform. Note that a single person can be multiple users.

**Post:** Stores relevant information about posts; they must be posted by a valid user. A user cannot make multiple posts at the exact same time.

**Repost:** Stores repost details. They must be tied to valid posts and users.

**Project:** Stores relevant details about projects. Projects must have unique names.

**Field:** Stores the fields of a project. A field is associated with its project via the foreign key project\_name.

**Project\_Post:** Stores the association between projects and posts. The primary keys of both tables are stored as foreign keys here, allowing for a many to many association.

**FieldResult:** Stores field results for posts in projects. It follows the same association conventions used above.

**Note:** Capitalization in the actual tables differs from this schema.

## Instruction Manual

### 1. Setup

- a. Install [Node.JS](#). Default setup suffices if using the installer.
- b. Clone this repository: `git clone https://github.com/QuinnMatthews/CS5330-DB-Project.git`
- c. Navigate into the `social-media-research-database` folder (`cd .\CS5330-DB-Project\social-media-research-database`)
- d. Install MySQL
  - i. Windows
    1. Go to <https://dev.mysql.com/downloads/installer/> and download the installer
    2. Choose Developer Default. The rest of the configuration should be straightforward.
    3. Use the command: `mysql -u root -p` to verify your installation.
  - ii. Mac
    1. In the terminal, perform the following:

```
brew update
brew install mysql
brew services start mysql
mysql_secure_installation
mysql -u root
```
  - iii. Linux
    1. In the shell, perform the following:

```
sudo apt update
sudo apt install mysql-server
sudo mysql_secure_installation
sudo mysql
```
- e. Create the database for the project:  
`mysql -u root -p` (or `mysql -u root` or `sudo mysql` depending on your OS)  
`CREATE DATABASE dbproj;`
- f. Copy the `.example.env` file to `.env` and modify the values as appropriate.
- g. Run `npm install` to install the necessary packages.
- h. Run `npm run dev` to start the program.
- i. Go to `localhost:3000`.

### 2. Usage

- a. Left Sidebar (Navigation)
  - i. Home
  - ii. Manage Platforms
  - iii. Manage Users
  - iv. Manage Posts
  - v. Manage Projects
  - vi. Search Posts

- vii. Search Experiments (Projects)
  - b. Home Page (Navigation)
    - i. Home
    - ii. Manage Platforms
    - iii. Manage Users
    - iv. Manage Posts
    - v. Manage Projects
    - vi. Search Posts
    - vii. Search Experiments (Projects)
  - c. Manage Platforms
    - i. Platform Creation
      - 1. Platform Name (Required)
      - 2. Add Platform Button
    - ii. Existing Platform Management
      - 1. Platform Name (Not an action)
      - 2. Delete Button
  - d. Manage Users
    - i. Add New User Button
      - 1. Choose Platform (Required)
      - 2. Username (Required)
      - 3. Cancel
      - 4. Add User
    - ii. Edit User Menu
      - 1. First Name (Optional)
      - 2. Last Name (Optional)
      - 3. Birthdate (Optional)
      - 4. Gender (Optional)
      - 5. Birth Country (Optional)
      - 6. Verified (Optional)
      - 7. Cancel
      - 8. Save Changes
    - iii. Delete User
  - e. Manage Posts
    - i. Add New Post
      - 1. Select Platform (Required)
      - 2. Select Username (Required)
      - 3. Post Text (Required)
      - 4. Seconds Known (Has Default Value)
      - 5. Date/Time Posted (Required)
      - 6. City (Optional)
      - 7. State (Optional)
      - 8. County (Optional)
      - 9. Likes (Has Default Value)
      - 10. Dislikes (Has Default Value)

11. Includes Multimedia (Has Default Value)
12. Cancel
13. Add Post
- ii. Filter
  1. Platform (Optional)
  2. Username (Optional)
  3. First Name (Optional)
  4. Last Name (Optional)
  5. Start Date (Optional)
  6. End Date (Optional)
  7. Search
- iii. Posts
  1. Reposts
    - a. New Repost
      - i. User (Required)
      - ii. Date/Time (Required)
      - iii. Seconds Known (Has Default Value)
      - iv. Add Repost
    - b. Reposts
      - i. Delete Repost
    - c. Close
  2. Edit
    - a. Post Text (Required)
    - b. City (Optional)
    - c. State (Optional)
    - d. Country (Optional)
    - e. Likes (Has Default Value)
    - f. Dislikes (Has Default Value)
    - g. Includes Multimedia (Has Default Value)
    - h. Cancel
    - i. Save Changes
  3. Delete
- f. Manage Projects
  - i. New Project
    1. Project Name (Required)
    2. Institute (Required)
    3. Manager First Name (Required)
    4. Manager Last Name (Required)
    5. Start Date (Required)
    6. End Date (Required)
    7. Add Initial Fields (Optional)
      - a. New Field Name (Required)
      - b. Add
    8. Manage Initial Fields

- a. Field Name (Not Modifiable)
    - b. Remove
  - ii. Projects
    - 1. Edit
      - a. Institute (Required - if missing, modification does nothing)
      - b. Manager First Name (Required - if missing, modification does nothing)
      - c. Manager Last Name (Required - if missing, modification does nothing)
      - d. Start Date (Has Default Value)
      - e. End Date (Has Default Value)
      - f. Save Changes
      - g. Manage Fields
        - i. Add Field
          - 1. Field Name (Required)
          - 2. Add
        - ii. Delete Field
    - 2. Manage Posts
      - a. Add New Post
        - i. Platform (Required)
        - ii. Username (Required)
        - iii. Datetime (Required)
        - iv. Seconds Known (Has Default Value)
        - v. Text (Required)
        - vi. City (Optional)
        - vii. Region (Optional)
        - viii. Country (Optional)
        - ix. Likes (Has Default Value)
        - x. Dislikes (Has Default Value)
        - xi. Multimedia (Has Default Value)
      - b. Link Existing Post
        - i. Search Posts (Dynamic Search)
        - ii. Link
      - c. Edit Results
        - i. Field 1 (Optional)
        - ii. ... (Optional)
        - iii. Field N (Optional)
        - iv. Update Results
      - d. Unassociate
    - 3. Delete
  - g. Search Posts
    - i. Platform (Optional)
    - ii. Username (Optional)
    - iii. First Name (Optional)

- iv. Last Name (Optional)
  - v. Start Date (Optional)
  - vi. End Date (Optional)
  - vii. Search
- h. Search Experiments (Projects)
  - i. Select Project (Required)